1. Lead 16.4: the free automorphism fold of the Gaudry base over $\mathbb{F}_{p^2}$

Round 4, 2026-10-11. Example `examples/r4_16_4_zeta_fold_fp2.rs`. Raw records in this directory: `census_torsion.json`, `pipeline_{prime,torsion}.json`, `stream_{prime,torsion}.json`, `sweep_torsion_{10,12}.json` (plus `.txt` logs). Full tables, exact commands and limits are in `RESULTS.md`. Toy synthetic curves only. No claim about rho, sub-rho behaviour or deployed curves. Wall times are contended.

1.1. Question and arms

On $E : y^2 = x^3 + b$ over $\mathbb{F}_{p^2}$, $p \equiv 1\ 3$, the Weil-descent base $F = \{P : x(P) \in \mathbb{F}_p^*\}$ is closed under $\zeta(x, y) = (\omega x, y)$. Prop. 15.2 predicts: (b) folding by $\langle -1, \zeta \rangle$ instead of (a) $\langle -1 \rangle$ costs $\times\ 1/3$, and (c) a base saturated by a rational 3-torsion point T is neutral at $m = 2$ and loses $\times\ 3$ at $m = 3$. Arm (c) is $F_c = \varphi^{-1}\{Q \in E' : x(Q) \in \mathbb{F}_p^*\}$ for the 3-isogeny $\varphi : E \to E' : y^2 = x^3 - 27b$ with kernel $\langle T \rangle$. It folds by $\langle -1, \zeta, \tau_T \rangle$ (18 points per column). Through φ it is the base $F'' = \{Q \in \varphi(E) : x(Q) \in \mathbb{F}_p^*\}$ on E'.

1.2. Structural finding (derived; checked on every instance)

Rational 3-torsion on $y^2 = x^3 + b$ requires b to be a square. Then Frobenius $\pi = \left(\zeta_6^j \pi_1\right)^2$ is a square, so E is a base change from $\mathbb{F}_p$ and $\#E = N_1 N_2$ with $N_i \approx p$. **The census shape $\#E = 3r$ of §15.2 cannot occur over $\mathbb{F}_{p^2}$.** Arm (c) exists only on curves with $r \le (\sqrt{p} + 1)^2$. So two families were used. The prime family (b a non-square, $h \le 16$, $r \approx p^2$) runs arms (a) and (b). The torsion family (b a square, $9 \nmid \#E$, $h \approx p$) runs all three arms on one curve.

1.3. Results

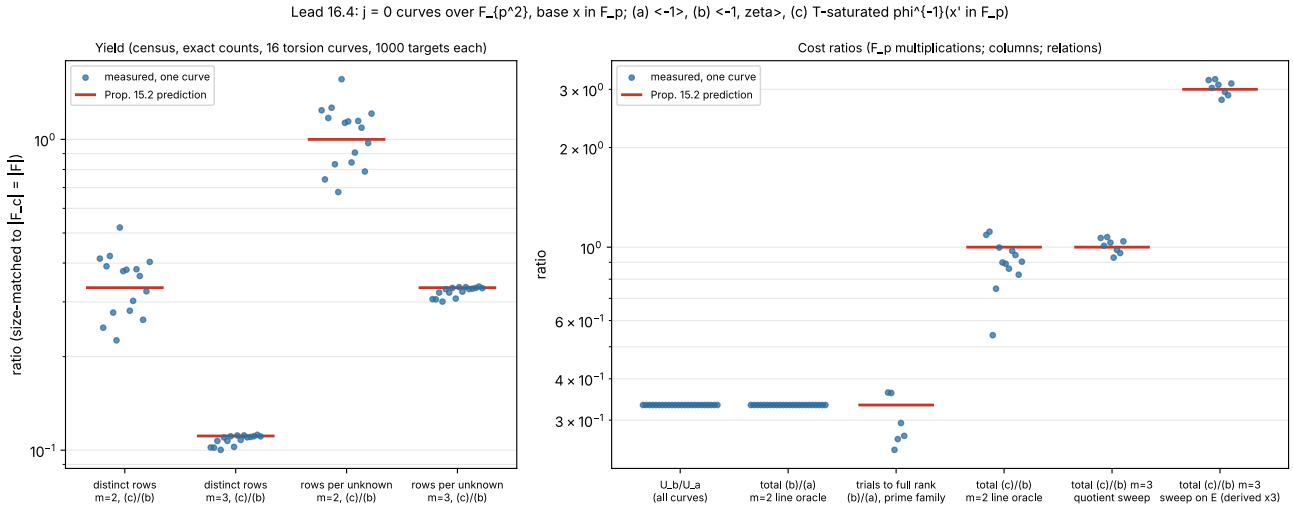


Figure 1: Recorded ratios, one point per curve, against the Prop. 15.2 prediction (red). Left: exact census yields (16 torsion curves, $p \approx 2^9$ to 2^{12}, 1000 targets each), size-matched to $|F_c| = |F|$. Right: cost ratios. U_b/U_a and total (b)/(a) come from the repository's `LineOracle` (24 curves, p up to 2^{16}). Trials to full rank come from `full_rank_stream` (6 prime-family curves). Total (c)/(b) at $m = 2$ uses the line oracle on 12 torsion curves. Total (c)/(b) at $m = 3$ uses the example's sweep oracle (8 curves); the "on E" column is derived as $3 \times$ the quotient sweep. Sources: the JSON files listed above.

comparison	law	measured (range over curves)	status
(b)/(a) total, $m = 2$, line oracle	1/3	0.3333 on 24 of 24 curves (6 points/column, strict ζ-closure)	confirmed, exact
(a)/(b) relations to full rank	3 × coupon factor	2.79–4.19 (6 streams, log recovered on all)	confirmed
(c)/(b) total, $m = 2$, line oracle, size-matched	1	0.54–1.11; 8 of 12 per-curve 95% intervals contain 1	confirmed (neutral; Poisson saturation predicts about 0.85)
(c)/(b) distinct rows, $m = 3$	1/9	0.100–0.112 (16 curves)	confirmed
(c)/(b) rows per unknown, $m = 3$	1/3	0.301–0.336 (16 curves)	confirmed
(c)/(b) total, $m = 3$, sweep over F_c on E	3	2.67–3.62 (derived: 3 × quotient sweep)	reproduces the law
(c)/(b) total, $m = 3$, quotient sweep on E'	3	0.93–1.07 size-matched; 95% intervals exclude 3 on 8 of 8	**law's ×3 refuted for this oracle**

1.4. Verdict

The ζ-fold is free and exact. The base is automatically ζ-closed, the columns drop by exactly 3, and the descent cost per call is unchanged. The repository supports it but had no driver for plain $j = 0$ curves over $\mathbb{F}_{p^2}$. The kernel-saturated arm follows the law's yield predictions exactly: the raw/distinct multiplicities are 3.00 and 9.0, and the size-matched yields are $1/3$ and $1/9$. Its $m = 3$ cost penalty of ×3 appears only for an oracle that enumerates over F_c. An algebraic oracle in the quotient coordinate $x' = x \circ \varphi$ enumerates $F_c/\langle T \rangle$, and there the arm is measured neutral (≈ 1). The $|M|^{m-2}$ term of Prop. 15.2 should therefore be stated per oracle.

Limits: toy sizes ($p \leq 2^{16}$; census and sweep $p \leq 2^{12}$). The $m = 3$ oracle is the example's sweep and returns one pair per call. Arm (c) runs only on base-change curves. Torsion-family streams did not reach full rank, though all recovered the logarithm. Inversions are not priced.